NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY KHULNA

CSE 4208: Pattern Recognition Lab

Discussion On CKD Using ML On Patients Clinical Records

Section: 8B | Group: 4

Presented By

Md. Mahfujul Karim Sheikh
ID:11220120735

Md Riaz Ahmed
ID:11220120740

Arnesha Sarker Shithi
ID:11220120734

Nasim Hossain
ID:11210320646

Md. Abdullha Al Mamun
ID:11180320272

Dept. Of CSE, Northern University of Business And Technology Khulna

Presented To

Md. Azizul Islam
Senior Lecturer &
Coordinator

Dept. Of CSE, Northern University of Business And Technology Khulna

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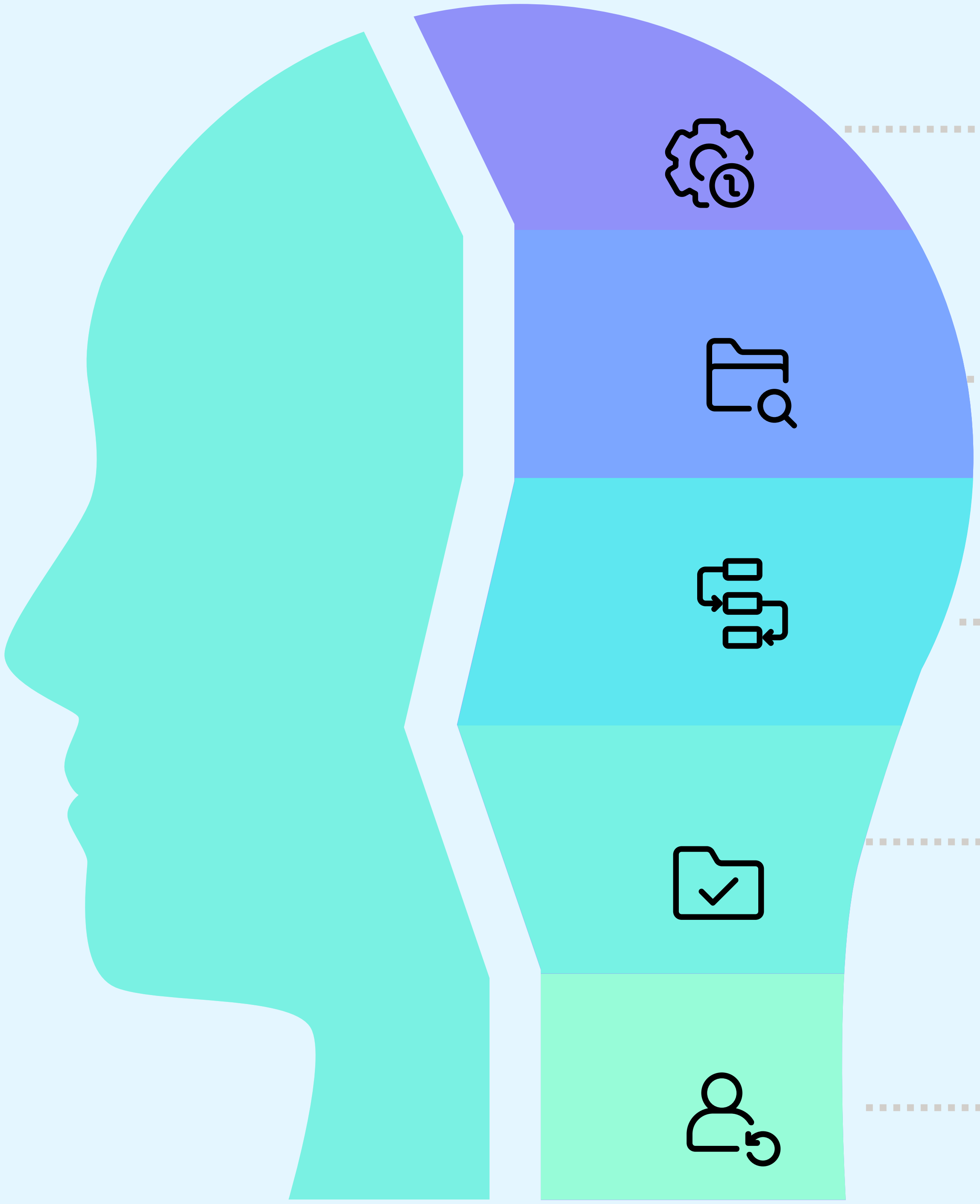
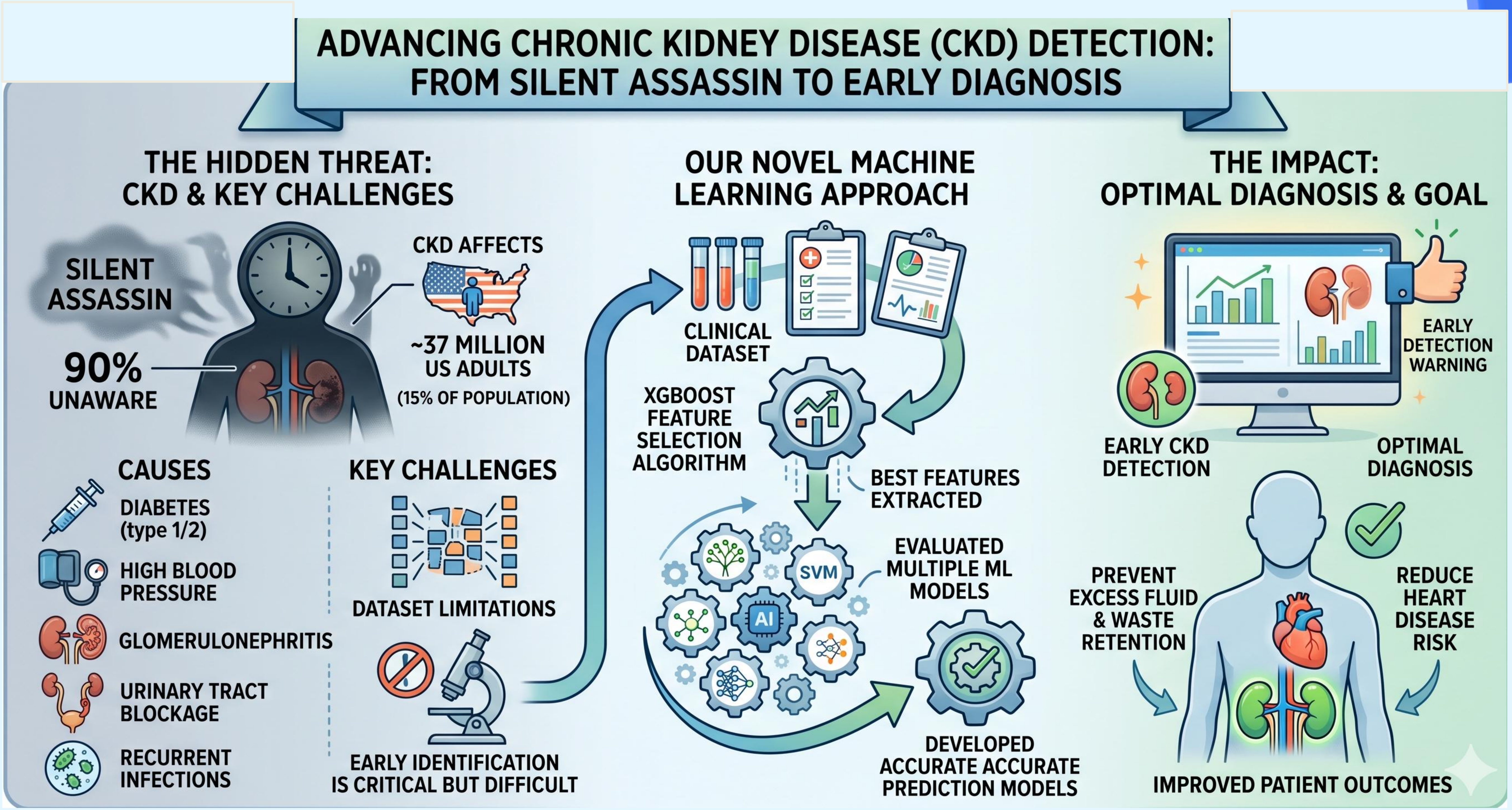


Introduction(Abstract)



Literature Review

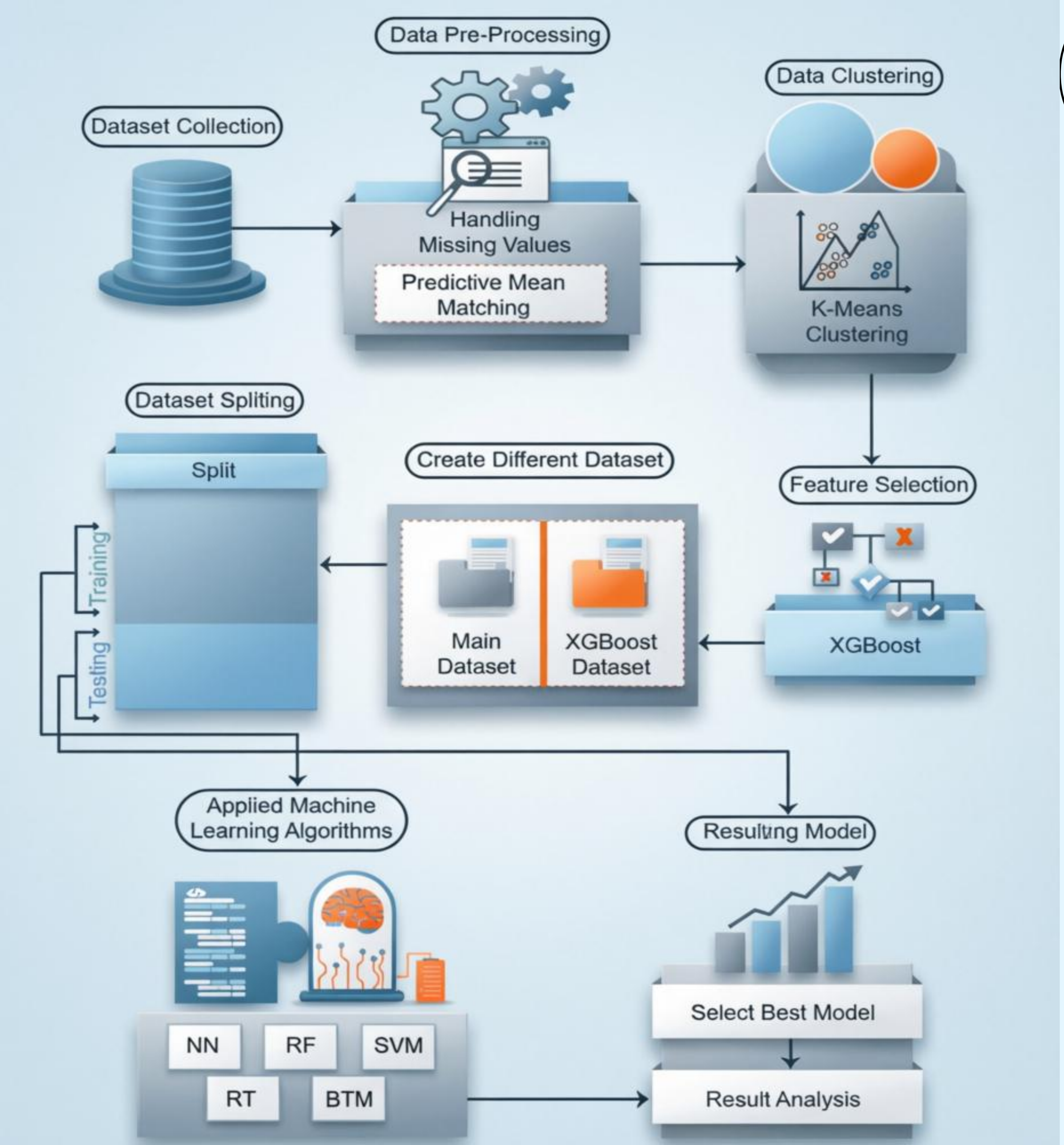
PREVIOUS WORK HIGHLIGHTS

Study	Method/Algorithm	Accuracy/Result
Alaiad et al.	NB, J48, SVM, KNN, JRip, Various	—
Sobrinho et al.	J48 best	—
Khan et al.	SVM	98.25%
Gunarathne et al.	Multclass Decision Forest	99.1%
Alasker et al.	DT all 24 features	98.41%
Abdullah et al.	RF + Feature Selection	98.825%
Charleonnnan et al.	SVM	97.3%

Better Feature Engineering → Fair Comparison of Multiple Algorithms → Higher Accuracy and Reliability

THE NEED FOR: COMPREHENSIVE FEATURE ENGINEERING AND COMPARISON

Methodology



Dataset Preprocessing(with description)

1. DATASET DESCRIPTION

Source: UCI Machine Learning Repository – CKD

STATISTICS

Total Records: 400 patients

Total Features: 25 clinical characteristics

Target: CKD / Non-CKD classification

FEATURE CATEGORIES

CONTINUOUS VARIABLES

- Age
- Blood Pressure
- BGR
- Blood Urea
- Serum Creatinine
- Sodium
- Potassium
- Hemoglobin
- PCV
- WBC
- RBC

BINARY VARIABLES

- Specific Gravity
- Albumin
- Sugar
- RBC
- Pus Cell
- Pus Cell Clumps
- Bacteria
- Hypertension
- Diabetes Mellitus
- Coronary Artery Disease (CAD)
- Appetite
- Pedal Edema

2. DATA PREPROCESSING

MISSING VALUE IMPUTATION – PREDICTIVE MEAN MATCHING (PMM)

WHY PMM?

- Ideal for quantitative non-normal variables
- Produces more plausible results than other methods
- Better represents diverse data patterns

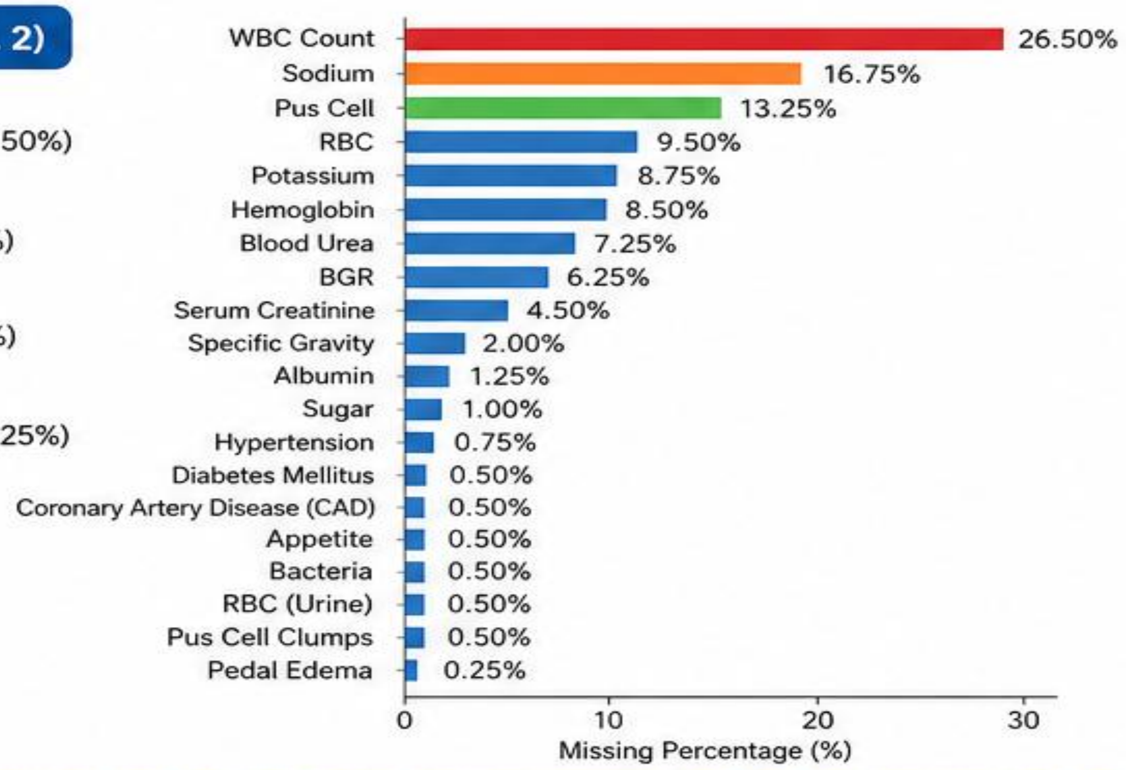
MISSING DATA ANALYSIS (Fig. 2)

Highest Missing: WBC count (26.50%)

Second Highest: Sodium (16.75%)

Third Highest: Pus Cell (13.25%)

Lowest Missing: Pedal Edema (0.25%)



PROCESSING STEPS

1. Strip whitespace from categorical variables
2. Replace '?' and 'None' with NaN
3. Encode categorical features (LabelEncoder)
4. Impute using IterativeImputer (BayesianRidge)

K-MEANS CLUSTERING ANALYSIS

OBJECTIVE

Identify distinct patient subgroups based on clinical characteristics.

METHOD

- StandardScaler normalization
- Elbow method for optimal k
- K-means clustering

RESULTS

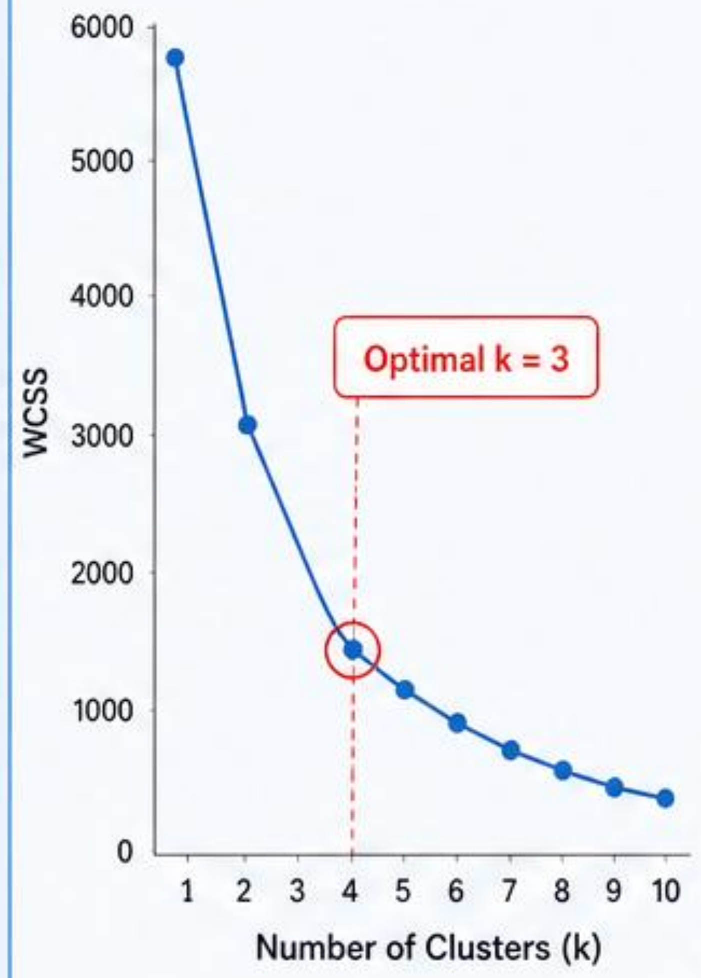
- Optimal k = **3 clusters**
- Clear separation in feature space
- Validated with dendrogram analysis

VISUALIZATIONS

- a Elbow Method – WCSS vs k
- b K-means Clustering (k = 3)
- c Hierarchical Dendrogram

a Elbow Method

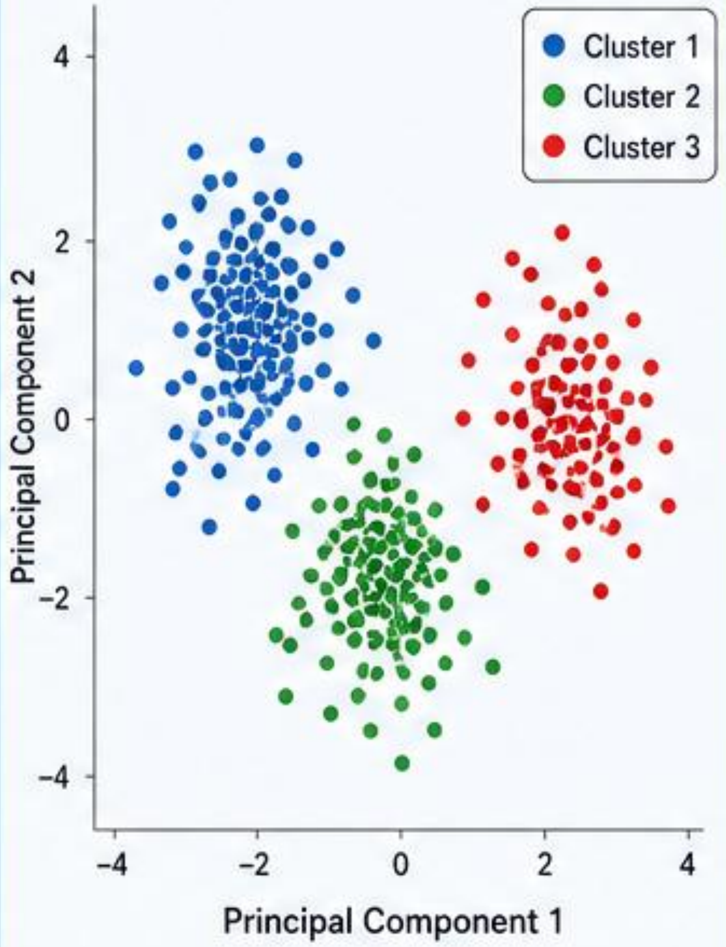
WCSS vs k



The elbow point at k = 3 suggests the optimal number of clusters.

b K-means Clustering (k = 3)

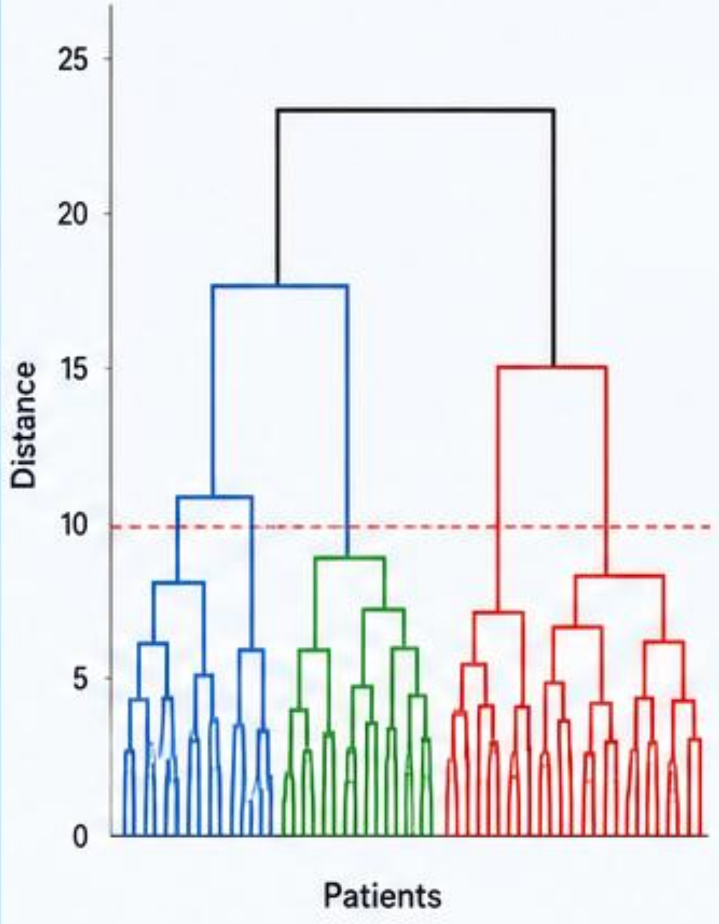
PCA-reduced Feature Space



Three distinct clusters with clear separation in the feature space.

c Hierarchical Dendrogram

Ward Linkage



Dendrogram cut at distance ≈ 10 confirms 3 natural clusters.

INSIGHT

Natural **patient groupings** exist in the clinical feature space.

Feature Selection

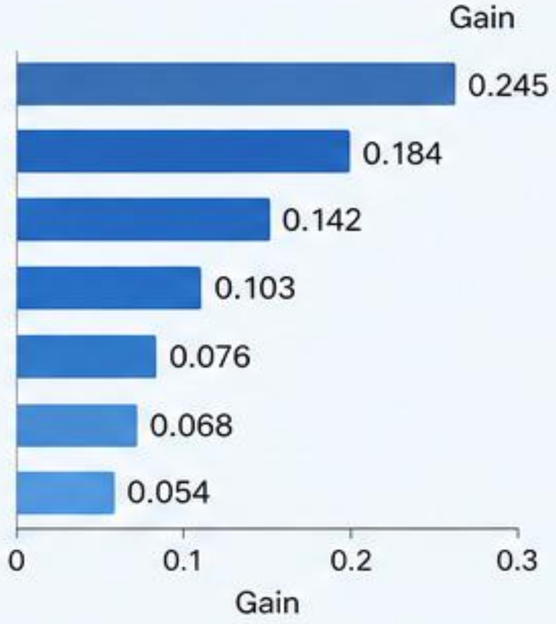
1. FEATURE SELECTION – XGBOOST

WHY XGBOOST?

- Gradient boosting ensemble method
- Handles feature importance naturally
- Reduces overfitting
- Improves model accuracy

TOP 7 FEATURES SELECTED
(by Gain)

1. Hemoglobin (hemo) – Most important
2. Serum Creatinine (sc)
3. Packed Cell Volume (pcv)
4. Albumin (al)
5. Pedal Edema (pe)
6. Blood Pressure (bp)
7. Specific Gravity (sg)



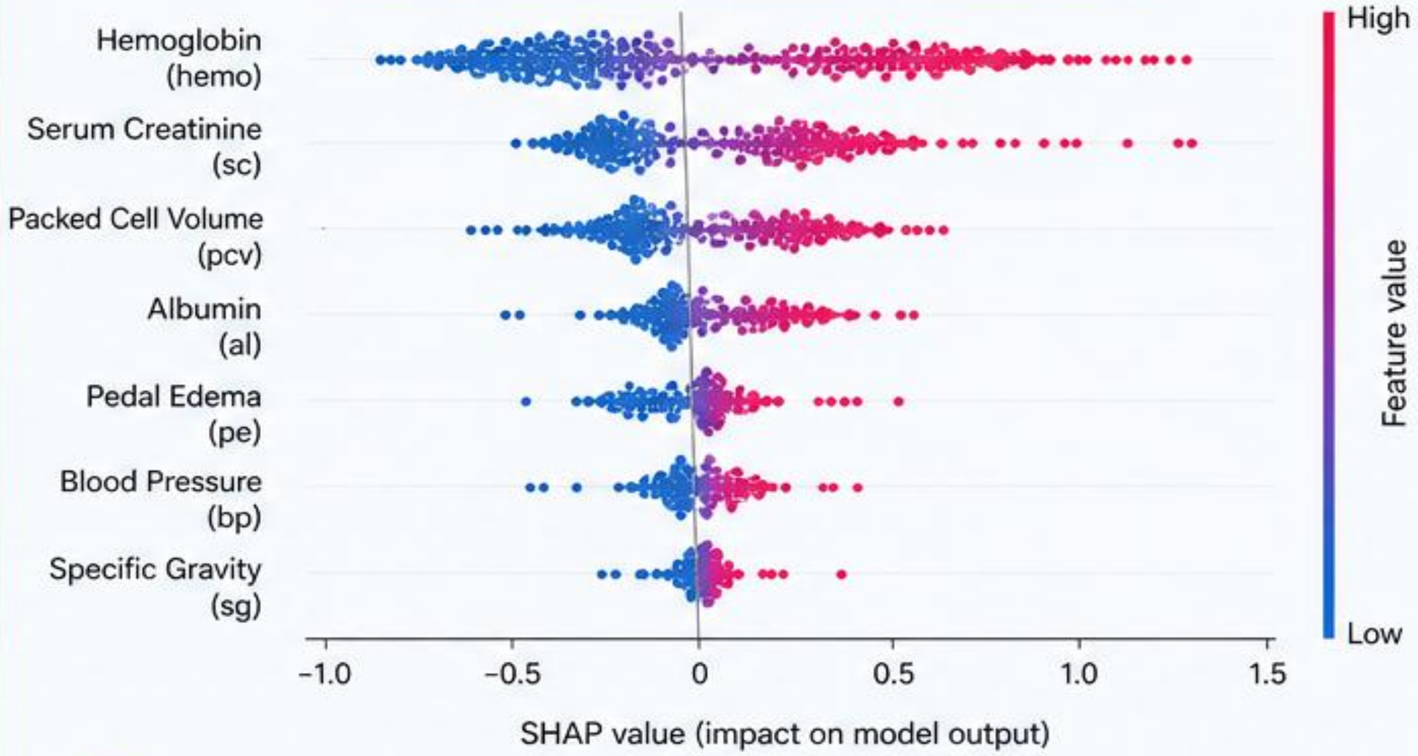
SHAP ANALYSIS

- Quantifies feature impact on predictions
- Shows direction and magnitude of impact
- Validates clinical relevance

SHAP

2. SHAP FEATURE IMPORTANCE

SHAP Summary Plot (Impact on Model Output)

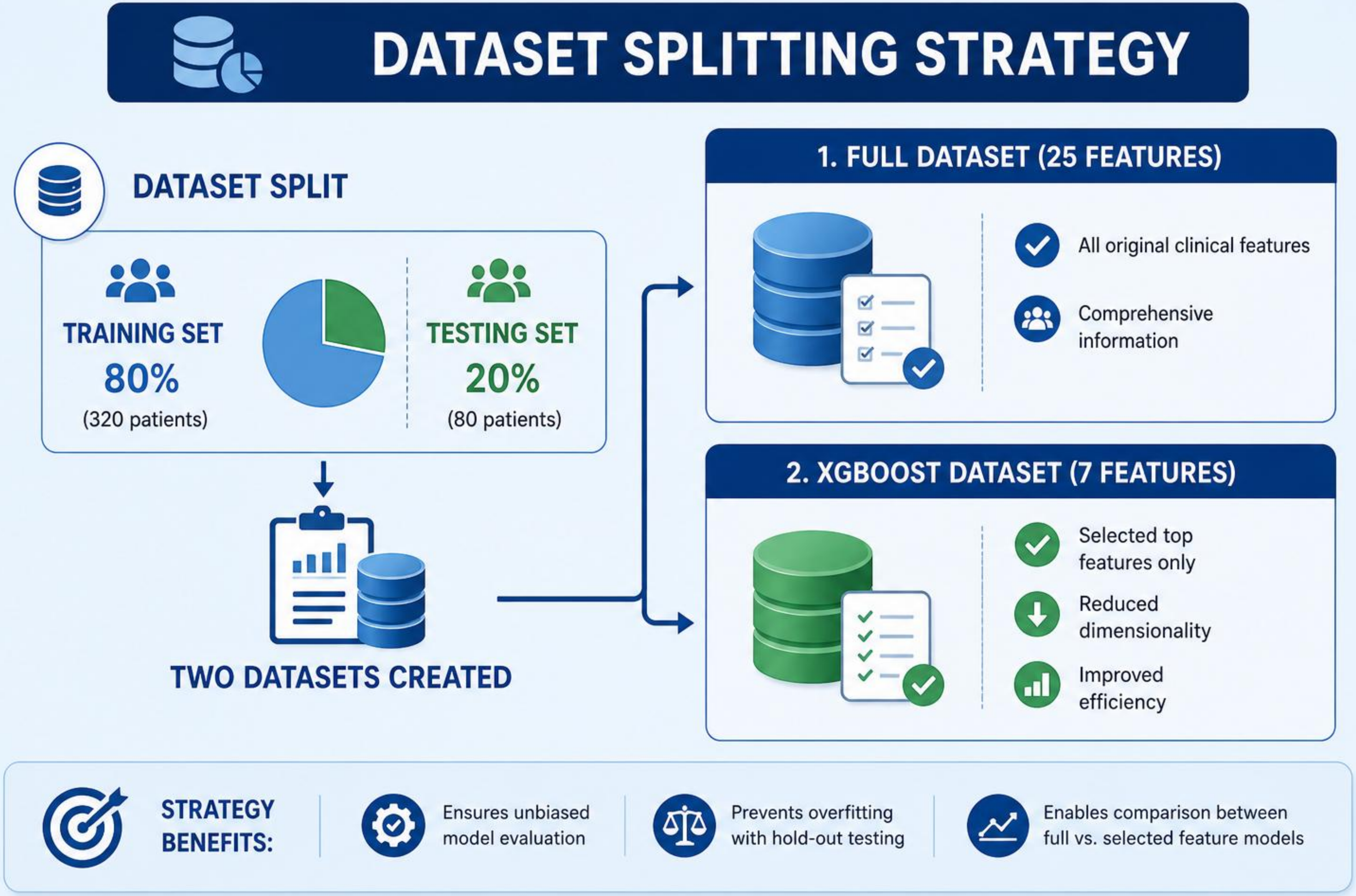


KEY FINDINGS FROM SHAP ANALYSIS

Hemoglobin (hemo):
- Highest overall impact on CKD prediction
- Lower values strongly indicate CKD
- Clinically validated biomarker

Serum Creatinine (sc):
- Second most important
- Direct kidney function indicator
- Strong positive SHAP values for higher risk

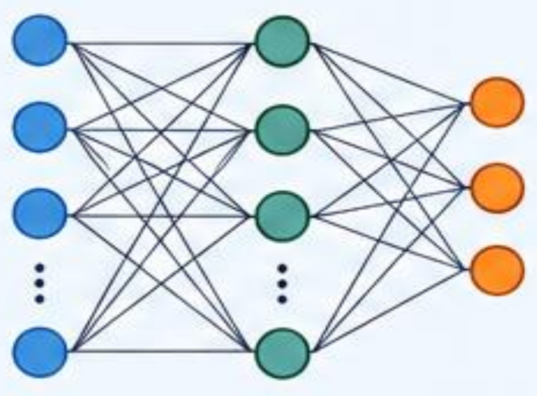
Pattern:
- Features align with known clinical knowledge
- Model captures medically relevant signals
- Supports model interpretability and trust


DATASET SPLITTING STRATEGY
DATASET SPLIT
TRAINING SET
80%
(320 patients)
TESTING SET
20%
(80 patients)
TWO DATASETS CREATED
1. FULL DATASET (25 FEATURES)
All original clinical features
Comprehensive information
2. XGBOOST DATASET (7 FEATURES)
Selected top features only
Reduced dimensionality
Improved efficiency
STRATEGY BENEFITS:
Ensures unbiased model evaluation
Prevents overfitting with hold-out testing
Enables comparison between full vs. selected feature models

MACHINE LEARNING MODELS

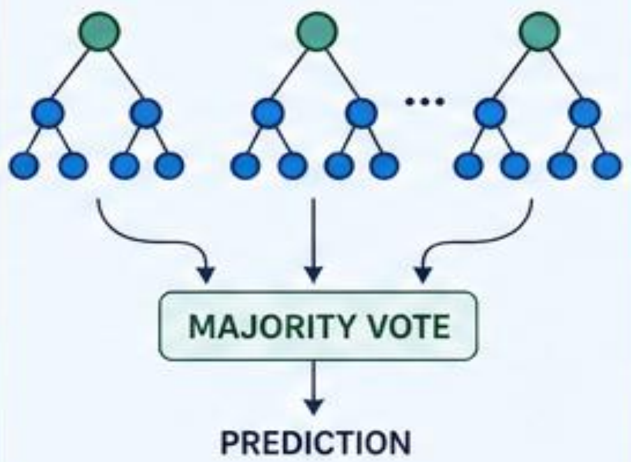
FIVE ALGORITHMS EVALUATED

1 NEURAL NETWORK (NN)



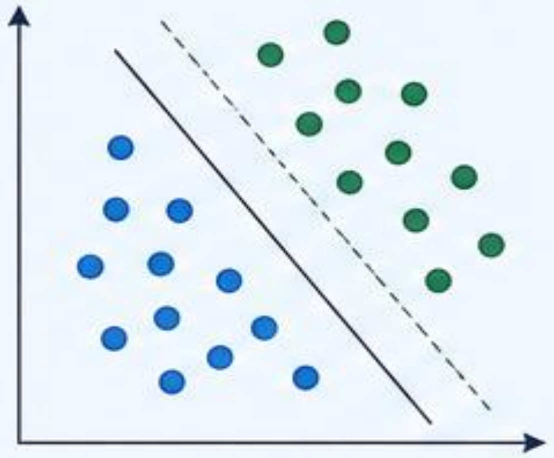
- Architecture: 64-32 hidden layers
- Early stopping enabled
- Scaled input features

2 RANDOM FOREST (RF)



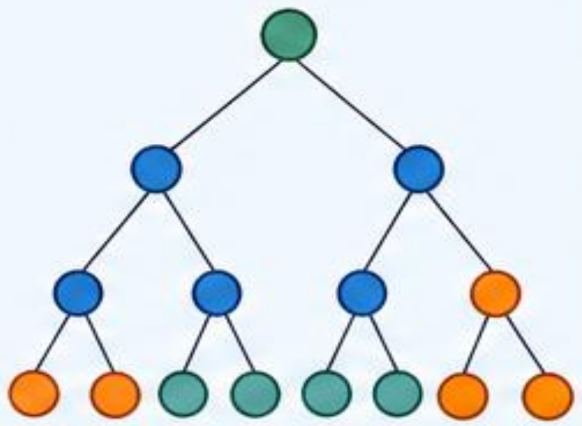
- 100 estimators
- Ensemble of decision trees
- Handles non-linearity well

3 SUPPORT VECTOR MACHINE (SVM)



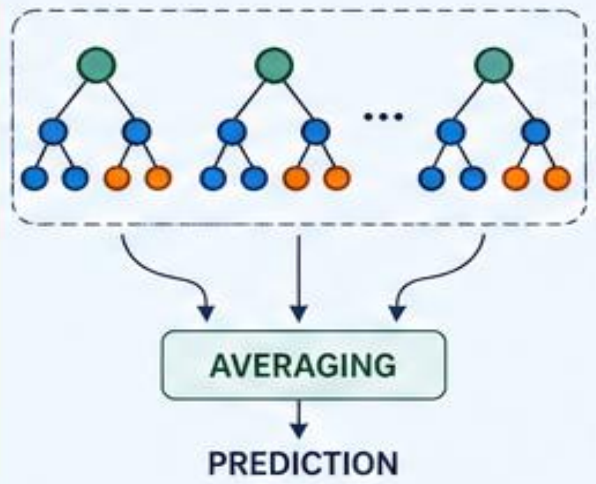
- RBF kernel
- C = 10, gamma = 'scale'
- Effective in high dimensions

4 RANDOM TREE (RT)



- ExtraTreesClassifier
- Extremely randomized trees
- Reduces overfitting

5 BAGGING TREE MODEL (BTM)



- DecisionTree base estimator
- 100 estimators
- Variance reduction

WHY THESE MODELS?

- Diverse learning approaches
- Captures complex patterns
- Improves prediction accuracy
- Robust & reliable performance
- Reduces overfitting and variance

RESULTS SUMMARY – TABLE 3

MODEL PERFORMANCE COMPARISON (Acc%, Sen%, Spe%, Kappa%)

Model	FULL DATASET (25 FEATURES)				XGBOOST DATASET (7 FEATURES)			
	Acc (%)	Sen (%)	Spe (%)	Kappa (%)	Acc (%)	Sen (%)	Spe (%)	Kappa (%)
NN	~97.50	~95.00	~100.00	~95.00	~97.50	~95.00	~100.00	~95.00
RF	~98.75	~97.50	~100.00	~97.50	~98.75	~97.50	~100.00	~97.50
SVM	~98.75	~97.50	~100.00	~97.50	**100.00**	**100.00**	**100.00**	**100.00**
RT	~96.25	~95.00	~97.50	~92.50	~96.25	~95.00	~97.50	~92.50
BTM	~96.25	~95.00	~97.50	~92.50	~97.50	~95.00	~100.00	~95.00

KEY FINDING: SVM achieves **100% accuracy** on XGBoost-selected features!

Highest Performance
SVM with XGBoost features achieved perfect scores across all metrics.

Consistent Results
RF also shows excellent performance across both datasets.

High Specificity
Most models achieved 100% specificity, minimizing false positives.

Model Reliability
High Kappa values indicate strong agreement and reliable predictions.

Acc: Accuracy, Sen: Sensitivity, Spe: Specificity, Kappa: Cohen's Kappa

COMPARISON WITH EXISTING WORKS

Table 4: Our Results vs Literature

Model	Our Accuracy (%)	Paper Accuracy (%)	Existing Work Accuracy (%)	Reference
NN	~97.50	100.00	99.75	Almansour et al. [26]
RF	~98.75	98.75	92.54	Hore et al. [28]
SVM	~98.75	98.75	98.25	Khan et al. [12]
RT	~96.25	96.25	95.50	Almustafa et al. [32]
BTM	~96.25	96.25	96.00	Hasan et al. [34]

KEY INSIGHTS

- Our models achieve competitive accuracy across all algorithms.
- RF shows the highest improvement over existing work.
- SVM and NN perform on par with the best reported results.
- Our results validate the effectiveness of feature selection and modeling approach.

Hands-On Experiment

[Mahfujul-01726/ckdcode]

Conclusion

PAPER FINDINGS VS OUR RESULTS

- SVM achieves **100%** accuracy on XGBoost-selected features
- NN achieves **~97.5%** accuracy on full 25-feature dataset
- All models exceed **96%** accuracy – robust performance
- Feature selection validated – maintains/improves results

CLINICAL IMPACT

- Early detection enables **timely intervention**
- Reduces progression to **end-stage renal disease**
- Lowers **healthcare costs**
- Improves **patient outcomes**

FUTURE WORK

- **Real-time** deployment in clinical settings
- **Integration** with electronic health records
- Multi-center **validation** studies
- **Deep learning** exploration

Our proposed approach demonstrates high accuracy, reliability, and *clinical relevance* for chronic kidney disease prediction.

Questions & Answers

Reference

PRIMARY SOURCE

Hassan, M.M., Hassan, M.H., Mollick, S., et al. (2023).
"A Comparative Study, Prediction and Development of Chronic Kidney Disease Using Machine Learning on Patients Clinical Records."
Human-Centric Intelligent Systems, 3:92–104.
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SUPPORTING LITERATURE

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  Deep neural network for chronic kidney disease prediction.
  IEEE Access, *9*, 12383–12393.
- Khan, A., Sharif, M., Raza, M., & Khan, S.U. (2020).
  Chronic kidney disease prediction using support vector machine.
  Computers in Biology and Medicine, 121, 103792.
- Gunarathne, T., Wijayarathna, S., & Perera, C. (2019).
  Decision forest methods for the prediction of chronic kidney disease.
  International Journal of Advanced Computer Science and Applications, 10(8), 102–110.
- Alasker, H., Alhajj, R., & Alharbi, A. (2018).
  Chronic kidney disease prediction using data mining classifiers.
  Journal of Medical Systems, *42*(7), 1–10.

DATASET

- UCI Machine Learning Repository: Chronic Kidney Disease Dataset
  https://archive.ics.uci.edu/ml/datasets/Chronic_Kidney_Disease

These references provide the foundation for our study and comparison with existing works in chronic kidney disease prediction using machine learning techniques.